Emergency Triage, Red Flags, Rapid Screen & Subjective

A · PATIENT DEMOGRAPHICS & INJURY PROFILE — INTAKE

Patient Name _____ **Age** _____ **Sex** ☐ M ☐ F **Date** _____

Sport / Occupation _____ **Level** ☐ Rec ☐ Comp ☐ Elite **Dominant leg** ☐ L ☐ R **Side** ☐ L ☐ R ☐ Bilat

Date of Injury / Surgery _____ **Referrer** _____ **Prior knee injury / surgery** _____ **Imaging** _____

Mechanism ☐ Non-contact pivot / deceleration (valgus–ER) ☐ Hyperextension ☐ Dashboard (posterior tibial force) ☐ Direct valgus / varus blow ☐ Twist on loaded flexed knee ☐ Patellar dislocation ☐ Insidious / overuse

At injury ☐ Audible "pop" ☐ **Swelling < 2 h** (haemarthrosis: ACL / fracture / patellar dislocation) ☐ Swelling 6–24 h (meniscal) ☐ Unable to continue

NPRS Now (circle) 0 1 2 3 4 5 6 7 8 9 10 **Worst 24 h** _____ **Best 24 h** _____ **Activity** _____

B · EMERGENCY SCREENING & RED FLAGS — COMPLETE BEFORE ANY LOADING OR MANUAL THERAPY — TRIAGE GATE

OTTAWA NOT APPLICABLE: ☐ Injury > 7 days ☐ Pregnancy ☐ Altered consciousness ☐ Paraplegia ☐ Multiple / distracting injuries ☐ Isolated skin injury
· Age < 18: derived in adults; paediatric validation Sn 100% (Bulloch 2003)

OTTAWA KNEE RULE → RADIOGRAPH IF ANY [H] — MDCalc — Sn 98.5% · Sp 49%

- ☐ Age ≥ 55
- ☐ Tenderness at **head of fibula**
- ☐ **Isolated patellar** tenderness
- ☐ Unable to **flex to 90°**
- ☐ **Unable to bear weight 4 steps** (immediately & at assessment)

☐ Negative → no X-ray ☐ Positive → knee radiograph

☐ **Pittsburgh** [M] (alternative): blunt trauma / fall + age < 12 or > 50 **or** unable 4 WB steps — Sn 99% · Sp 60%

HIGH-RISK INJURY CHECKLIST — Do not miss

- ☐ **Knee dislocation / multiligament** (≥ 2 ligaments; may self-reduce): pulses + **ABI** _____ → **ABI < 0.90 = arterial injury → CTA / vascular** [M] · check common peroneal n. (foot drop)
- ☐ **Extensor mechanism rupture** — unable to SLR / extensor lag + palpable gap · patella alta (tendon) or baja (quadriceps) → urgent orthopaedic
- ☐ **Segond fracture** (lateral tibial rim avulsion = ACL tear until proven otherwise) • **Tibial plateau Fx** (axial + valgus; lipohaemarthrosis)
- ☐ **Patellar dislocation + haemarthrosis** → imaging for osteochondral fracture / loose body
- ☐ **SKELETALLY IMMATURE?** → **tibial spine avulsion** (mimics ACL) • physeal / tibial tubercle injury → X-ray

NON-ORTHOPAEDIC & REFERRED RED FLAGS — Systemic

- ☐ **Wells DVT** (post-op / immobilised / calf swelling): ☐ ≥ 2 → duplex US ☐ < 2 unlikely Score _____ MDCalc
- ☐ **Compartment syndrome** (plateau Fx / crush / post-op) ⓘ : ☐ Pain on passive stretch ☐ Paraesthesia ☐ Pallor ☐ Paralysis ☐ Pulseless
  ΔP (Diastolic BP − Compartment Pressure) ≤ 30 mmHg = fasciotomy. PT: recognise & refer now.
- ☐ **Septic arthritis** — hot effusion · fever · unable to WB · recent injection / arthroscopy / arthroplasty · immunosuppressed → **ED same day (aspiration)**
- ☐ **Crystal arthritis** — gout / pseudogout (CPPD: older adult, knee commonest site) → urgent medical review
- ☐ **Bone tumour** — distal femur / proximal tibia (osteosarcoma peak 10–25 y) · **unremitting night pain** · mass · weight loss · cancer history
- ☐ **SIFK / osteonecrosis** — > 50 y, sudden severe medial pain without trauma, night pain → MRI, protect weight-bearing
- ☐ **Referred from the HIP** — adolescent knee/thigh pain + limp + ↓ hip IR / obligatory ER on hip flexion → **SCFE: urgent hip X-ray (AP + frog-leg), non-weight-bearing**
- ☐ **Popliteal artery** — pulsatile popliteal mass (aneurysm) or exertional calf claudication in a young athlete (entrapment)
- ☐ **Budapest CRPS** [H] Calc ≥ 3/4 Sx + ≥ 2/4 signs: ☐ Sensory ☐ Vasomotor ☐ Oedema / sudomotor ☐ Motor / trophic

TRIAGE DISPOSITION: ☐ **Cleared — proceed** ☐ Imaging referral ☐ Urgent orthopaedic ☐ **Vascular / Emergency Department NOW**

C · PATHWAY ROUTER & FAST-TRACK ASSESSMENT (< 2-MINUTE SCREEN) — RAPID

CLINICAL PATHWAY ROUTER

Tick the route → tap a page number

- ☐ **A · SIMPLE:** 1 → 6 → 7 → 9
- ☐ **B · COMPLEX:** 1 → 2 → 3 → 4 → 6 → 7 → 8 → 9
- ☐ **C · POST-OP:** 1 → 5 → 6 → 7 → 8 → 9

Effusion — stroke test ▸ [M]
☐ 0 ☐ T ☐ 1+ ☐ 2+ ☐ 3+

Active SLR / extensor lag
☐ No lag ☐ Lag _____ °

Knee ROM (°)
Ext _____ Flex _____

Single-leg squat × 5 ▸ [M]
☐ Good ☐ Fair ☐ Poor

Fast Classification: ☐ Patellofemoral pain (atraumatic) ☐ MCL sprain Gr I ☐ Early patellar tendinopathy ☐ Stable knee OA

D · 24-HOUR SYMPTOM BEHAVIOUR, PAIN MAP & PATIENT GOALS — SUBJECTIVE

MECHANICAL SYMPTOMS

- ☐ **True locking** (block to extension) → displaced bucket-handle tear: urgent orthopaedic
- ☐ Catching / clicking
- ☐ Giving way: ☐ pivot (instability) ☐ pain inhibition
- ☐ Swelling after activity

PAIN PATTERN

- ☐ **Anterior**: squat / stairs / prolonged sitting → PFP
- ☐ **Inferior pole**, load-dependent, warms up → tendinopathy
- ☐ Activity pain, AM stiffness ≤ 30 min, age ≥ 45 → OA
- ☐ **AM stiffness > 30 min / multiple joints** → inflammatory

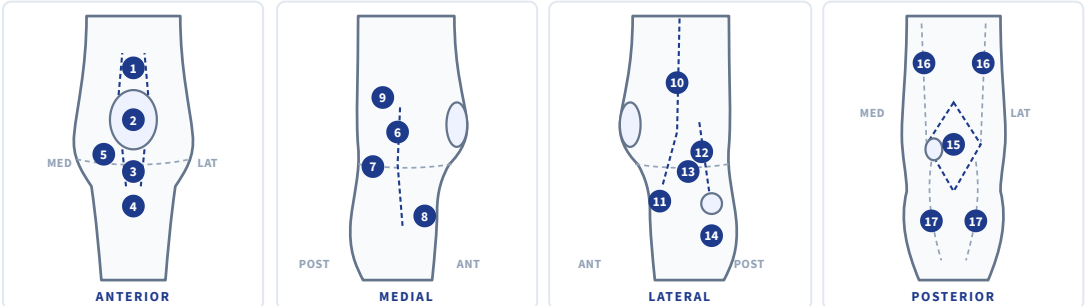
Night pain ☐ Mechanical ☐ **Unremitting**

AGGRAVATING

☐ Stairs ↑ ☐ Stairs ↓ ☐ Squat ☐ Kneeling ☐ Sitting > 30 min ☐ Running ☐ Jumping ☐ Cutting / pivot ☐ Sit-to-stand

PMH / Risk: ☐ Prior ACL / meniscal surgery ☐ Inflammatory arthritis ☐ Diabetes ☐ BMI ≥ 30 ☐ Fluoroquinolones ☐ Steroid injection ☐ Hypermobility ☐ Anticoagulants ☐ Smoker

PAIN MAP — shade symptoms · X = max tenderness · circle side: L R · numbers = palpation targets, p3



1 Quadriceps tendon · **2** Patella / retinacula · **3** Inferior pole / patellar tendon · **4** Tibial tubercle · **5** Infrapatellar fat pad · **6** MCL · **7** Medial joint line · **8** Pes anserine · **9** Adductor tubercle / MPFL · **10** ITB / lat. epicondyle · **11** Gerdy's tubercle · **12** LCL · **13** Lateral joint line · **14** Fibular head / CPN · **15** Popliteal fossa / Baker's · **16** Hamstring tendons · **17** Gastrocnemius heads

GOALS — PSFS — MDC 2

1. _____ /10
2. _____ /10
3. _____ /10

Expectation _____

Timeline _____

Clinician _____ **Date** _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

Unbiased Differential Diagnosis & Neurological Matrix

Patient _____

Bias-free decision architecture: screen **every** compartment before committing to a hypothesis. Tick all **suspected** conditions, then rank the top 3 in the Working Hypothesis box. Mark L/R where relevant. Hip and lumbar spine are screened on every knee patient (bottom card).

1 · ANTERIOR KNEE — 7 TARGETS

- ☐ **Patellofemoral pain (PFP)** — retro/peripatellar pain on squat, stairs, prolonged sitting; diagnosis of exclusion
- ☐ **Patellar tendinopathy** ▸ — inferior pole, load-dependent (jump/energy storage) • **Quadriceps tendinopathy** — superior pole
- ☐ **Patellar instability** — subluxation / dislocation history; (+) moving apprehension test ▸; J-sign; Beighton
- ☐ **Infrapatellar fat pad (Hoffa)** ▸ — pain in terminal extension / hyperextension, either side of patellar tendon
- ☐ **Osgood-Schlatter** (tibial tubercle) • **Sinding-Larsen-Johansson** (inferior pole) — adolescent apophysitis
- ☐ **Prepatellar / infrapatellar bursitis** — superficial fluctuant swelling, kneeling; **warm + fever → septic (p1)**
- ☐ **Medial plica** — painful click, tender band medial to patella

2 · MEDIAL KNEE — 5 TARGETS

- ☐ **MCL sprain** ▸ — valgus stress 30° (isolated) / 0° (laxity = combined injury): ☐ Gr I ☐ Gr II ☐ Gr III
- ☐ **Medial meniscus tear** — joint line tenderness ▸, McMurray ▸, Thessaly ▸ → composite p4
- ☐ **Pes anserine bursitis / tendinopathy** — 2–5 cm below medial joint line, often with OA
- ☐ **Medial tibiofemoral OA** — age ≥ 45, activity-related pain, AM stiffness ≤ 30 min
- ☐ **Saphenous nerve** — burning medial knee / leg; Tinel at adductor canal; (+) saphenous neurodynamic test

3 · LATERAL KNEE — 5 TARGETS

- ☐ **ITB syndrome** — runners / cyclists, pain ~30° flexion at lateral epicondyle; (+) Noble compression ▸, Ober ▸
- ☐ **LCL / posterolateral corner** — varus stress 0° / 30° ▸; dial test ▸ > 10° side-to-side
- ☐ **Lateral meniscus tear** — lateral joint line tenderness, Thessaly, McMurray
- ☐ **Popliteus tendinopathy** (downhill running) • **Proximal tibiofibular joint** instability
- ☐ **Common peroneal nerve** — Tinel at fibular neck, dorsum-foot numbness, **foot drop**

4 · POSTERIOR KNEE — 5 TARGETS

- ☐ **Baker's (popliteal) cyst** — posteromedial swelling, often with OA / meniscal tear; **rupture mimics DVT → duplex US**
- ☐ **PCL injury** — dashboard / fall on flexed knee; (+) posterior sag ▸, posterior drawer ▸
- ☐ **Distal hamstring / biceps femoris tendinopathy** — pain on resisted flexion, insertion tenderness
- ☐ **Medial gastrocnemius strain** ("tennis leg") — sudden calf pain on push-off
- ☐ **Popliteal artery** aneurysm / entrapment, **DVT** → vascular red flags p1

5 · INTRA-ARTICULAR & GLOBAL — 6 TARGETS

- ☐ **ACL rupture** ▸ — pivot mechanism, "pop", haemarthrosis < 2 h, giving way → cluster p4
- ☐ **Meniscal tear** — catching / locking, delayed effusion (6–24 h) → composite p4
- ☐ **Osteochondral lesion / OCD** — adolescent, vague pain, effusion, catching; Wilson test [L]
- ☐ **Patellofemoral / tibiofemoral OA** — crepitus, bony enlargement, no warmth (ACR p4)
- ☐ **Inflammatory / crystal arthritis** — warm effusion, AM stiffness > 30 min → p1
- ☐ **Post-op arthrofibrosis** — persistent extension loss, ↓ patellar mobility, patella infera

WORKING HYPOTHESIS RANKING — SYNTHESIS

1° _____ Conf. ____ %

2° _____ Conf. ____ %

3° _____ Conf. ____ %

Must-not-miss excluded? ☐ Yes ☐ No

Irritability ☐ Low ☐ Mod ☐ High

Mechanism ☐ Nociceptive ☐ Neuropathic ☐ Nociplastic

Imaging ☐ Not needed ☐ X-ray ☐ MRI ☐ US

SPINAL, HIP & KINETIC CHAIN SCREENING — LUMBAR L2–S1 NEURO-MAPPING — NEURO

LEVEL	MYOTOME (KEY MUSCLE)	REFLEX	DERMATOME	MYOTOME L / R (0–5)	REFLEX L / R (0–4+)	SENSATION L / R
L2	Iliopsoas (hip flexion)	—	Anterior upper thigh	L ____ R ____	L ____ R ____	L ____ R ____
L3	Quadriceps (knee extension)	Patellar (L3–L4)	Anterior thigh → medial knee	L ____ R ____	L ____ R ____	L ____ R ____
L4	Tibialis anterior (DF / inversion)	Patellar	Medial leg & medial malleolus	L ____ R ____	L ____ R ____	L ____ R ____
L5	Extensor hallucis longus	Medial hamstring	Dorsum foot / 1st web space	L ____ R ____	L ____ R ____	L ____ R ____
S1	Gastrocnemius & peroneals	Achilles	Lateral foot & sole	L ____ R ____	L ____ R ____	L ____ R ____

Grading key: myotome ≤ 4/5 = deficit · reflex 0–1+ hypo / ≥ 3+ hyper · sensation intact / impaired / absent

NEURAL TENSION & DISTAL BIASING

Positive = familiar Sx + structural differentiation

TEST	SEQUENCE / BIAS	ONSET ANGLE L / R (°)	FAMILIAR SX	STRUCTURAL DIFF. (+)
Prone Knee Bend — Femoral ▸	Prone knee flexion (+ hip extension); L2–L4 / anterior thigh	L ____ R ____	☐	☐ slump-PKB neck flex
Slump — Sciatic / tibial ▸	Thoracic + cervical flexion, knee ext, ankle DF → release neck	L ____ R ____	☐	☐ Sx ↓ neck ext
Saphenous bias	Prone: hip ext + abd, knee ext, ankle DF + eversion	L ____ R ____	☐	☐
Common peroneal bias	SLR + ankle plantarflexion + inversion	L ____ R ____	☐	☐

Hip & chain: ☐ Hip IR loss / pain ☐ FADIR / FABER ▸ reproduces Sx ☐ Lumbar ROM reproduces Sx ☐ Hip abductor weakness ☐ Dynamic knee valgus ☐ Ankle DF deficit

⛔ ☐ **HIP REFERRAL — adolescent with knee pain + limp + ↓ hip IR = SCFE until proven otherwise (urgent hip X-ray, non-weight-bearing). Adults: hip OA often presents as knee pain — always examine the hip.**

DIFFERENTIAL REASONING NOTES · IMAGING FINDINGS

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

QUANTITATIVE MOBILITY & JOINT STATUS — LEFT VS RIGHT

ROM · EFFUSION · ALIGNMENT

MEASURE	NORMATIVE / THRESHOLD	LEFT	RIGHT	Δ	FLAG / INTERPRETATION
Knee extension (°)	0° to −5° hyperextension · **any deficit vs other side matters** E				☐ Extension deficit
Prone heel-height difference (cm)	≈ 1 cm → 1° extension loss E				☐ > 1 cm
Knee flexion (°)	135–150° · gait ≈ 65° · stairs ≈ 90–100° · squat ≥ 115° E				☐ Functional limit
Effusion — stroke test ▸	0 / trace / 1+ / 2+ / 3+ · inter-rater κ 0.75 M				☐ ≥ 1+ → reduce load
Patellar mobility (quadrants)	Med / lat glide 1–2 quadrants · sup / inf glide symmetric E				☐ Hypo ☐ Hyper (≥ 3)
Thigh girth 10 / 15 cm above patella (cm)	Side-to-side difference = quadriceps atrophy				☐ Atrophy
Hip IR, prone (°)	Compare sides · loss + groin pain → screen hip (p2)				☐ Restricted
Hip extension — modified Thomas (°)	Thigh on table, knee ≈ 80° flexed (rectus femoris) L				☐ Iliopsoas ☐ RF
Hamstring 90/90 AKE (° from full ext)	Compare sides; deficit = hamstring length L				☐ Short
Ankle DF — WBLT (cm)	≥ 10 cm · asymmetry > 1.5 cm (knee valgus driver) M				☐ Restricted
Beighton score (/9) ▸ Calc	Hypermobile: ≥ 5 adults to 50 y · ≥ 6 children · ≥ 4 over 50 y M	/ 9			☐ Hypermobile
Static alignment	☐ Genu varum ☐ Genu valgum ☐ Recurvatum ☐ Patella alta ☐ Squinting patellae ☐ Pronated foot (FPI-6 _____)				

Goniometry: measurement error ≈ 3–5° between sessions — differences smaller than this are not meaningful.

MUSCLE CAPACITY — HAND-HELD DYNAMOMETRY (N · NM/KG)

HHD · LSI

Body mass _____ kg Lever arm _____ m Torque (Nm/kg) = Force (N) × lever arm (m) ÷ body mass (kg) · strap-fixed HHD for quadriceps M

MUSCLE	TEST POSITION	L (N)	R (N)	L NM/KG	R NM/KG	LSI %
Quadriceps	Seated, knee 60–90°, strap to frame, 3 × 5 s MVIC					
Hamstrings	Prone or seated, knee 90°					
Hip abductors	Side-lying, hip neutral, strap above knee					
Hip external rotators	Seated, hip & knee 90°					
Hip extensors	Prone, knee 90°					

Quadriceps activation: ☐ Extensor lag on SLR ☐ No superior patellar glide on quad set ☐ Arthrogenic inhibition (AMI) suspected → NMES / cryotherapy p6

FUNCTIONAL PERFORMANCE

OARSI · SPORT

TEST	RESULT	RE-TEST
30-s chair stand (reps) H		
40-m fast-paced walk (s → m/s) H		
Stair climb test (s) H		
Timed Up & Go (s)		
Single-leg squat × 5 ▸ M	☐ Good ☐ Fair ☐ Poor	
Fault: ☐ Trunk lean ☐ Pelvic drop ☐ Hip adduction / IR ☐ Knee valgus		

OARSI-recommended core set for knee OA: 30-s chair stand, 40-m fast-paced walk, stair climb.

SYSTEMATIC PALPATION — 17 TARGETS (MAP P1)

TICK TENDER SIDE

ANTERIOR	L · R	LATERAL	L · R
1. Quadriceps tendon	☐☐	10. ITB / lat. epicondyle	☐☐
2. Patella / retinacula	☐☐	11. Gerdy's tubercle	☐☐
3. Inferior pole / pat. tendon	☐☐	12. LCL	☐☐
4. Tibial tubercle	☐☐	13. Lateral joint line	☐☐
5. Infrapatellar fat pad	☐☐	14. Fibular head / CPN	☐☐
MEDIAL		**POSTERIOR**	
6. MCL (origin → insertion)	☐☐	15. Popliteal fossa / Baker's	☐☐
7. Medial joint line	☐☐	16. Hamstring tendons	☐☐
8. Pes anserine	☐☐	17. Gastrocnemius heads	☐☐
9. Adductor tubercle / MPFL	☐☐		

Warmth ☐ L ☐ R **Crepitus** ☐ PFJ ☐ TFJ

DYNAMIC MOVEMENT QUALITY

GAIT · STEP-DOWN · LANDING

TASK	OBSERVED FAULTS (TICK)	L	R
Gait	☐ Antalgic ☐ Stiff-knee / quad avoidance ☐ No terminal extension at heel strike ☐ Varus / valgus thrust		
Lateral step-down (0–6) M	☐ Arm strategy ☐ Trunk lean ☐ Pelvis not level ☐ Knee medial to 2nd toe (1) / foot (2) ☐ Unsteady 0–1 good · 2–3 moderate · ≥ 4 poor		
Drop vertical jump	☐ Knee valgus on landing ☐ Stiff landing (little knee flexion) ☐ Side-to-side asymmetry → full LESS scoring p8		

GAIT · OBSERVATION · MOVEMENT QUALITY NOTES

Evidence-Based Special Test Clusters & Validated PROMs

Patient _____

How to read: statistics belong to the exact test or cluster named — never transfer a cluster value to a single test. Sn rules **out** when negative; Sp rules **in** when positive. · Interactive: underline = Atlas card · ▸ = video · Calc = calculator · Guide = open-access guideline

ACL RUPTURE — INSTABILITY

History: ☐ Non-contact pivot ☐ "Pop" ☐ Haemarthrosis < 2 h ☐ Giving way

- ☐ **Lachman** ▸ — Sn 85% · Sp 94% [H]
- ☐ **Pivot shift** ▸ rule-in; often – when guarding — Sn 24% · Sp 98% [H]
- ☐ **Anterior drawer** ▸ chronic only; unreliable acute — Sn 92% · Sp 91% [M]
- ☐ **Lever sign** ▸ — Sn 81% · Sp 78% · –LR 0.21 [M]

☐ ACL suspected → MRI / orthopaedic ☐ Not supported

MENISCAL TEAR — COMPOSITE — 5 ITEMS

- ☐ Catching / locking history
- ☐ Pain on maximal passive flexion
- ☐ Joint line tenderness ▸
- ☐ Pain on forced hyperextension
- ☐ McMurray ▸ pain / click

Total ____ / 5

5/5 PPV 92% · ≥ 3/5 PPV > 75% (no ACL / OA) [M]

Single tests (pooled): JLT **Sn 83% Sp 83%** · McMurray **Sn 61% Sp 84%** · Thessaly 20° ▸ **Sn 75% Sp 87%** [M]

COLLATERALS & POSTEROLATERAL CORNER — VALGUS · VARUS · DIAL

- ☐ Valgus 30° ▸ → MCL
- ☐ Varus 30° ▸ → LCL
- ☐ Dial test ▸: > 10° ER side-to-side at **30° only** → PLC · at **30° and 90°** → PLC + PCL [E]
- ☐ Valgus 0° laxity → **combined**
- ☐ Varus 0° laxity → **PLC + cruciate**

Grade (joint opening): I ≤ 5 mm firm end-point · II 5–10 mm · III > 10 mm, no end-point [E]

PCL — POSTERIOR

☐ Posterior sag ▸ ☐ Posterior drawer ▸ ☐ Quad active ▸

Clinical exam overall Sn 90% · Sp 99% [M] posterior drawer = most accurate single test

Grade by medial tibial step-off at 90°: I still anterior · II flush with condyles · III posterior (suspect combined) [E]

PATELLOFEMORAL PAIN & INSTABILITY — PFJ

PFP — ALL THREE REQUIRED [M]

- ☐ Retro- or peripatellar pain
- ☐ Reproduced by ≥ 1 PFJ-loading task: ☐ Squat (most sensitive) ☐ Stairs ☐ Prolonged sitting ☐ Kneeling / jumping
- ☐ Other conditions excluded (p2)

INSTABILITY

- ☐ Moving patellar apprehension ▸ — Sn 100% · Sp 88% [L]

☐ J-sign ☐ Beighton ≥ 5 (p3) ☐ Prior dislocation

PATELLAR TENDINOPATHY & KNEE OA — LOAD · DEGENERATIVE

PATELLAR TENDINOPATHY ▸

☐ Inferior-pole pain ☐ Load-dependent ☐ SL decline squat ____ /10

Palpation tenderness alone has low specificity (common in asymptomatic athletes) [M]

KNEE OA — ACR CLINICAL: PAIN + ≥ 3 OF 6 — Sn 95% · Sp 69% [M]

☐ Age > 50 ☐ Bony tenderness ☐ Stiffness < 30 min ☐ Bony enlargement ☐ Crepitus ☐ No palpable warmth

NICE: ≥ 45 y + activity-related pain + no / ≤ 30 min AM stiffness = **clinical diagnosis, no imaging needed** [H]

ADDITIONAL INDIVIDUAL TESTS — RESULT LOG — L · R · (+) / (–)

TEST	L	R	TEST	L	R	TEST	L	R
Noble compression ▸ [E]			Ober ▸ [L]			Hoffa (fat pad) ▸ [E]		
Wilson (OCD) [L]			Prone knee bend ▸			Stroke test grade ▸		
Patellar tilt			Apley grind ▸ [L]					

STANDARDISED PATIENT-REPORTED OUTCOME MEASURES (PROMS) DASHBOARD — MCID · TRACKING

MEASURE	RANGE	MCID / MIC	BASE	WK 4	WK 8	WK 12	D/C	≥ MCID
KOOS — 5 SUBSCALES, 100 = NO PROBLEMS — KOOS Calc								
KOOS Pain	0–100	8–10 [M]						☐
KOOS Symptoms	0–100	8–10 [M]						☐
KOOS ADL	0–100	8–10 [M]						☐
KOOS Sport / Rec	0–100	8–10 [M]						☐
KOOS QOL	0–100	8–10 [M]						☐
IKDC subjective Calc	0–100	11.5 (6 mo) [M]						☐
Kujala / AKPS (PFP) Calc	0–100	10 [M]						☐
VISA-P (tendinopathy)	0–100	track trend						☐
Oxford Knee Score (OA / TKA) Calc	0–48	7 indiv. · 5 group [H]						☐
ACL-RSI (psychological) Calc	0–100	no universal cut-off						☐
Tegner activity level	0–10	pre-injury ____						☐

Non-responder rule: < MCID by week 6 → re-screen p1 red flags, reconsider the diagnosis, consider imaging or referral.

CLINICAL DIAGNOSIS · ICD-10 · PROGNOSIS & SMART GOALS

Clinician _____ Date _____

Criterion-Based Post-Surgical Protocols

Patient _____ 05 / 09

UNIVERSAL PROGRESSION GATE (every phase): ☐ Wound healed, no infection signs ☐ Pain ≤ 3/10 at rest ☐ Effusion ≤ 1+ and no rise after session (stroke test ▸) ☐ Full passive extension ☐ No DVT signs (Wells p1)
VTE prophylaxis: ☐ Prescribed ☐ Not indicated ☐ Surgeon-directed · re-screen Wells every visit while immobilised · **surgeon-specific restrictions override this template; time frames are minimums — progress on criteria**

1 · ACL RECONSTRUCTION — CRITERION-BASED PHASES

Aspetar CPG 2023

☐ BPTB ☐ HS ☐ QT ☐ ALLO ☐ + MENISCAL REPAIR

PHASE	GOALS & KEY INTERVENTIONS	CRITERIA TO PROGRESS	MET
1 · Protection 0–2 wk	Full passive extension = other side · quad activation: **NMES daily** [M] · patellar mobilisation · cryo-compression · WBAT with crutches (unless meniscal repair)	Full extension · SLR without lag · effusion ≤ 1+ · flexion ≥ 90°	☐
2 · Early strength 2–6 wk	Normal gait · CKC strength · **OKC knee extension 90→45° from week 4** (monitor anterior knee pain) [M] · low-load BFR optional (p8) [M]	Normal gait without aids · flexion ≥ 120° · effusion ≤ trace	☐
3 · Strength 6–12 wk	Progressive heavy strength (quads, hamstrings, hip, calf) · single-leg control · widen OKC range as symptoms allow	**Return-to-running gate p8:** ≥ 12 wk · full ROM · quad LSI > 80%	☐
4 · Power 3–6 mo	Running progression · plyometrics · change of direction · landing technique	Quad & hop LSI ≥ 90% · good landing quality (LESS)	☐
5 · Return to sport ≥ 9 mo	On-field rehabilitation · sport-specific load · psychological readiness	Pass p8 RTS battery · ACL-RSI rising · ≥ 9 months [M]	☐
Graft notes	**BPTB:** anterior knee / kneeling pain → patellar tendon load management · **HS:** avoid heavy isolated resisted knee flexion early (≈ 6–8 wk) [E] · **QT:** marked quadriceps inhibition → prioritise NMES & quad loading		

RED LINE: no pivoting, cutting or jumping before criteria. Reinjury risk falls 51% for each month RTS is delayed, up to 9 months (Grindem 2016).

2 · MENISCUS — MENISCECTOMY VS REPAIR

OJSM 2025

APM · REPAIR

☐ APM ☐ Repair: longitudinal / bucket ☐ Repair: radial / root

PARAMETER	PARTIAL MENISCECTOMY	REPAIR	MET
Weight-bearing	As tolerated from day 1	Longitudinal: WBAT in brace (extension) · **radial / root: NWB–TTWB 4–6 wk**	☐
ROM	Full as tolerated	0–90° for 4–6 wk, then progress	☐
Loaded flexion	As tolerated	**No deep squat / loaded flexion > 90° ≈ 12–16 wk**	☐
Running	≈ 4–6 wk (criteria)	≈ 12–16 wk (criteria)	☐
Return to sport	≈ 6–8 wk	≈ 4–6 months	☐

Typical ranges [E]; early WBAT after isolated repair shows comparable failure rates in recent reviews — follow the surgeon.

3 · TOTAL KNEE ARTHROPLASTY

TKA CPG 2020

TKA

PARAMETER	TARGET / INTERVENTION	MET
Start & mobilise	PT within 24 h · out of bed day 0–1 · wean gait aid on criteria	☐
ROM	Ext 0° · flex ≥ 90° early → ≥ 110–120° by 6–12 wk · **no CPM** (primary TKA) [H]	☐
Quadriceps NMES	From 48 h: 15 contractions, 2×/day, max tolerable, 6 wk [H]	☐
Strength & function	High-intensity strength from week 1 + balance / gait-symmetry training · cryotherapy early [H]	☐
Measures	KOOS JR · 30-s sit-to-stand · TUG (first & last visit)	☐
Flag	**Flexion < 90° at 6 wk → inform surgeon** (manipulation window) [E]	☐

4 · MPFL RECONSTRUCTION (PATELLAR STABILISATION)

IJSPT 2025 review

PATELLAR INSTABILITY

PHASE	GOALS & KEY INTERVENTIONS	CRITERIA TO PROGRESS	MET
Protection 0–6 wk	Brace / WB per surgeon (full WB 0–8 wk) · flexion from 0–30°, +20–30°/wk · **no lateral patellar glides** · quad activation	Full extension · SLR no lag · effusion ≤ 1+	☐
Strength 6–12 wk	Full flexion (by 4–12 wk) · CKC → OKC strength · hip & trunk control	Normal gait · full ROM · no apprehension	☐
Advanced 12–16 wk	Running, agility ladder, double-leg plyometrics, valgus-control landing	Quad LSI > 80% · hop LSI > 80%	☐
RTS > 16 wk	Cutting / pivoting drills → sport · no consensus RTS criteria — use p8 battery [L]	Quad & hop LSI ≥ 90% · confidence	☐

5 · CARTILAGE REPAIR (MICROFRACTURE / MACI)

CONDYLE · PFJ

PARAMETER	FEMORAL CONDYLE / PLATEAU	PATELLOFEMORAL LESION	MET
Weight-bearing	**Protected (TTWB–PWB) ≈ 6 wk**, full by 8–12 wk	WBAT in brace locked in extension	☐
ROM	Early full passive ROM encouraged	Flexion limited early (≈ 0–40/60°), progress	☐
Loading	CKC then OKC as graft matures	Avoid loading through the lesion contact arc	☐
Running / RTS	Running ≈ 6–9 mo · impact sport ≈ 9–18 mo (MACI longer than microfracture)		☐

Typical ranges [E] — lesion size, site and technique change everything: follow the operative note.

6 · QUADRICEPS / PATELLAR TENDON REPAIR

EXTENSOR MECHANISM

PARAMETER	TYPICAL PROGRESSION	MET
Brace / WB	WBAT in brace **locked in extension** ≈ 6 wk	☐
ROM	0–30° (wk 0–2) → 0–60° (2–4) → 0–90° (4–6)	☐
Quadriceps	Isometrics only; **no active / resisted extension ≈ 6 wk**	☐
Strength → RTS	Progressive strength 6–12 wk · running ≈ 4–5 mo · RTS ≥ 6 mo	☐

Typical ranges [E] — surgeon protocol overrides.

SURGEON-SPECIFIC INSTRUCTIONS · OPERATIVE FINDINGS · PHASE LOG

Clinician _____ Date _____

Underline = Atlas card · ▸ = video · Evidence: [H] high [M] moderate [L] low [E] expert · Confidential · v2026.10

EXTRACORPOREAL SHOCKWAVE THERAPY (ESWT) · ISMST Indications · 3–5 SESSIONS · WEEKLY

RX	INDICATION	ISMST STATUS	TYPE	ENERGY	PULSES	FREQ	TECHNIQUE
☐	**Patellar tendinopathy** M	Approved standard	fESWT	**0.10–0.30 mJ/mm²**	1500–2000	4–8 Hz	Knee 30–60° flexed; focus on inferior pole tenderness
☐	**Patellar tendinopathy**	Approved standard	rESWT	**2.0–3.0 bar**	2000–3000	8–12 Hz	Sweep proximal tendon; avoid patella bone impact
☐	**Pes anserine tendinopathy** L	Empirically tested	rESWT	**1.5–2.5 bar**	2000	8–10 Hz	Anteromedial tibia, below joint line
☐	**Knee osteoarthritis** L	Empirically tested	r / fESWT	**2.0–2.5 bar**	2000–4000	8–10 Hz	Tender joint-line & periarticular points
☐	**Osgood-Schlatter** L	Empirically tested	rESWT	**1.5–2.0 bar**	1500–2000	8 Hz	Low energy only over the apophysis

Energy / pulse ranges are typical clinical settings E; combine with progressive tendon loading (p8). ISMST contraindications: malignant tumour, fetus or pacemaker in the shockwave field.

DRUG & LOAD INTERLOCK: avoid routine NSAIDs during the loading phase where possible (mechanistic concern, evidence limited — prescribing rests with the physician). No ballistic loading for 48 h after ESWT. FLUOROQUINOLONE (current or within 6 months) or recent tendon steroid injection → NO ESWT and no heavy tendon loading.

PHOTOBIOMODULATION (LASER) · BMJ Open 2019 · EYEWEAR

RX	INDICATION	λ	DOSE PER SPOT	EV.
☐	**Knee OA**	785–860 nm	**4–8 J**	M
☐	**Knee OA**	904 nm	**1–3 J**	M
☐	**Tendinopathy / post-op**	810–980 nm	WALT ranges	L

6–10 spots along joint lines & popliteal fossa, 2–3×/wk × 2–4 wk E. Meta-analysis at these doses: ≈ 19 mm VAS less pain than placebo — **but NICE 2022 does not recommend laser for OA**: adjunct only, discuss with patient.
CI: malignancy in field, pregnancy (abdomen), photosensitising drugs, tattoos, eyes; Class IV (HILT): thermal risk.

TECAR THERAPY (448 KHZ) · CET · RET

MODE	ELECTRODE	TARGET TISSUE
CET	Insulated (capacitive)	Muscle, oedema, superficial
RET	Conductive (resistive)	Tendon, capsule, ligament

RX	LEVEL	INTENSITY	PHASE
☐	**Athermic**	5–15%	Acute 0–72 h
☐	**Mild thermal**	20–35%	Subacute
☐	**Hyperthermic**	55–75%	Chronic / stiffness

Evidence L · CI: pacemaker, pregnancy, DVT, malignancy, sensory loss, **metal implant in field (thermal modes, e.g. TKA)**.

CRYOTHERAPY & COMPRESSION · POST-OP · AMI

Post-op / acute: cryo-compression or ice 15–20 min, barrier, every 2–3 h · **Arthrogenic muscle inhibition (AMI):** cryotherapy 20 min **then** quadriceps exercise M · AMI review 2019
CI: cold urticaria, Raynaud's, cryoglobulinaemia, sensory loss, peripheral vascular disease, superficial peroneal / CPN pressure.

ELECTROTHERAPY — NMES · TENS · IFC · HVPC · EMG BIOFEEDBACK · QUADRICEPS FIRST

RX	GOAL	CURRENT	FREQUENCY	PULSE / CARRIER	INTENSITY	DOSE	EV.
☐	**Quadriceps strength — post-TKA**	NMES biphasic symmetric	30–50 Hz	250–400 µs	Max tolerable	15 contractions · 2×/day from 48 h · 10–15 s on / 45–50 s off · 6 wk	H
☐	**Quadriceps — post-ACLR / AMI**	NMES biphasic symmetric	30–50 Hz	250–400 µs	Max tolerable, with voluntary contraction	10–20 contractions daily, early phase · knee 60°, strap-resisted	M
☐	**Strength (alternative)**	Russian current	50 Hz bursts	Carrier 2500 Hz	Max tolerable	10 s on / 50 s off · 10–15 reps	L
☐	**Pain — acute / during exercise**	Conventional TENS	80–120 Hz	50–100 µs	Strong, comfortable, sub-motor	30–60 min; can run during exercise	L
☐	**Pain — chronic**	Acupuncture-like TENS	2–4 Hz	150–250 µs	Visible rhythmic contraction	20–30 min	L
☐	**Pain — deep tissue**	IFC quadripolar	Beat 80–150 / 1–10 Hz	Carrier 4000 Hz	Strong comfortable	15–20 min, sweep on	L
☐	**Oedema — acute**	HVPC (negative)	100–120 Hz	Twin-peak monophasic	Sub-motor	30 min, limb elevated	L
☐	**Oedema — calf muscle pump**	Motor-level NMES	1–10 Hz	200–300 µs	Rhythmic contraction	20–30 min, elevated	L
☐	**Quadriceps activation**	EMG biofeedback	Visual / auditory threshold on VMO during quad sets & SLR			3 × 10 contractions, every session	L

Quadriceps NMES set-up: electrode 1 over the VMO motor point (distal medial thigh) · electrode 2 over proximal lateral thigh (VL / RF) · knee 60° with strap resistance · intensity is the dose: aim for the strongest tolerable contraction. · **TENS:** intensity is the key parameter; identical daily settings produce tolerance within days — vary frequency.

Guideline limits: Knee OA — NICE 2022 does **not** recommend TENS, IFC, ultrasound, laser, pulsed short-wave or NMES (exercise, education & weight management first) H · **Patellofemoral pain** — electrophysical agents not recommended (2016 consensus) M · NMES after **TKA / ACLR** is a different indication and is recommended.

CI: pacemaker / ICD · over DVT · pregnancy (abdomen, low back) · active malignancy in field · broken skin · impaired sensation · epilepsy (head / neck) · transthoracic placement.

TREATMENT RESPONSE · ADVERSE REACTIONS · DOSAGE ADJUSTMENTS

Manual Therapy, Soft Tissue & Dry Needling

Patient _____

MANUAL THERAPY & NEURODYNAMICS

MAITLAND · MULLIGAN

RX	TECHNIQUE	TARGET	GRADE / DOSE
☐	**Tibiofemoral PA glide**	Extension ↓	III–IV, 3 × 30–60 s
☐	**Tibiofemoral AP glide**	Flexion ↓	III–IV, 3 × 30–60 s
☐	**Patellar glides (sup / inf / med)**	Post-op stiffness, patella infera	II–IV, 2–3 × 60 s
☐	**Long-axis distraction**	OA pain	I–II, 3 × 30–60 s
☐	**Proximal tibiofibular glide**	Lateral knee / PTFJ	II–IV, 3 × 30 s
☐	**MWM: tibial IR / ER glide**	Painful flexion / extension	3 × 10, pain-free only
☐	**Hip mobilisation**	Knee OA with hip signs	III–IV, 3 × 30–60 s [L]

NERVE	SLIDER	TENSIONER
Femoral (PKB / side-lying slump)	☐	☐
Saphenous	☐	☐
Sciatic / tibial (slump)	☐	☐
Common peroneal	☐	☐
Dose	2–3 × 10–15 (acute / irritable)	3 × 5–10 (chronic)

No lateral patellar glides after MPFL reconstruction or with lateral instability. Knee OA: manual therapy only **alongside** exercise [M]. **PFP: patellofemoral, knee & lumbar mobilisations not recommended** (2016 consensus) [M].

DRY NEEDLING SAFETY MAPPING

DEPTH VARIES WITH BMI · US GUIDANCE ADVISED

RX	MUSCLE	NEEDLE	DEPTH	APPROACH & NEUROVASCULAR SAFETY
☐	**Vastus lateralis**	0.30 × 50–60 mm	2–5 cm	Lateral thigh, perpendicular to femur; large safe window
☐	**Vastus medialis (VMO)**	0.30 × 40–50 mm	1.5–3 cm	Distal medial thigh; **stay anterior to sartorius** — femoral vessels in the adductor canal
☐	**Rectus femoris**	0.30 × 50–60 mm	2–4 cm	Anterior mid-thigh; **avoid the femoral triangle** proximally
☐	**Biceps femoris (distal)**	0.30 × 40–50 mm	1.5–3 cm	Lateral approach aimed laterally; **common peroneal nerve runs medial to the tendon**
☐	**Semitendinosus / membranosus**	0.30 × 40–60 mm	2–4 cm	Posteromedial thigh, aimed medially, away from the sciatic / tibial nerve
☐	**Gastrocnemius heads**	0.30 × 40–50 mm	1.5–3 cm	Pincer grip on each head; **never the centre of the popliteal fossa**
☐	**Popliteus (advanced)**	0.30 × 40–50 mm	2–3 cm	Posteromedial tibial border aimed laterally; popliteal artery, vein & tibial nerve lie centrally
☐	**TFL / gluteus medius**	0.30 × 60–75 mm	3–6 cm	Lateral-knee referral; perpendicular, superior to greater trochanter

Safety screen: ☐ Anticoagulants / bleeding screened ☐ No infection / lymphoedema ☐ Vasovagal history screened ☐ **No needling into the joint or near a knee replacement**
Knee OA: NICE 2022 — do not offer dry needling / acupuncture

MANUAL RELEASE & CUPPING

SOFT TISSUE

RX	TECHNIQUE	PARAMETERS	EV.
☐	**IASTM**	Quadriceps, TFL, hamstrings · 30–45° stroke angle · light–moderate pressure · 30–120 s per region · 1–2×/wk	L
☐	**Active release / pin-and-stretch**	Pin in shortened position → active lengthening · 3–5 passes	L
☐	**Foam rolling (self-release)**	Quadriceps / lateral thigh · 30–60 s × 3 · warm-up use; the ITB itself does not lengthen	L
☐	**Dry cupping — static**	Light–moderate suction · 5–10 min	L
☐	**Dynamic (sliding) cupping**	Lubricant · light suction · 3–5 min along fibre direction	L
☐	**Cupping with movement**	Cup in place + active knee ROM · 2–3 min	L
☐	**Wet cupping (Hijama)**	Skin-breaching: sterile single-use equipment, bloodborne-pathogen precautions. Confirm scope of practice & licensure before use.	L

CI: anticoagulants / bleeding disorders · fragile skin (elderly, chronic steroids) · over DVT / varicosities · open wounds or unhealed incisions · neuropathy · active infection · pregnancy (abdomen). Wet cupping also: anaemia, immunosuppression.

TAPING, BRACING & ORTHOSES

PFP Consensus 2016

ADJUNCTS

RX	DEVICE	INDICATION	RECOMMENDATION	EV.
☐	**Foot orthoses (prefabricated)**	Patellofemoral pain	Short-term pain relief; benefit varies between individuals	M
☐	**Patellar taping / bracing**	PFP, as part of combined care	Role on its own unclear — use only within exercise-based care	L
☐	**Unloader (valgus) brace / insoles**	Medial knee OA	Not routine (NICE): only with instability or abnormal loading when exercise alone is insufficient	M
☐	**Hinged post-operative brace**	ACLR / meniscal repair / MPFL / extensor repair	Surgeon-directed: lock angle & duration in operative note	E
☐	**Functional ACL brace**	Return to sport after ACLR	No clear reduction in reinjury — does not replace RTS criteria	L

PFP first line: combined hip + knee exercise [H] · **not recommended for PFP:** PF / knee / lumbar mobilisation, electrophysical agents [M]

CONSENT OBTAINED FOR: ☐ ESWT ☐ Dry needling ☐ Grade III–IV manual therapy ☐ Cupping (dry / wet) | ☐ Risks & alternatives discussed ☐ Patient may decline Signature _____

TECHNIQUE RESPONSE · ADVERSE REACTIONS

Clinician _____ Date _____

TENDON LOADING CONTINUUM — PATELLAR / QUADRICEPS TENDINOPATHY

BJSM 2021 RCT · TICK CURRENT STAGE

NOW	STAGE	PARAMETERS	PROGRESS WHEN
☐	**1 · Isometric**	Single-leg leg press or leg extension at 60° knee flexion · **5 × 45 s at ≈ 70% MVC** · daily	Single-leg squat pain **≤ 3/10** and ≥ 1 week at the stage
☐	**2 · Isotonic (heavy, slow)**	Leg press · split squat · leg extension · 3 s up / 3 s down · 15RM → 6RM over 12 wk (log below) · 3×/wk	
☐	**3 · Energy storage**	Jumps, hops, bounding · progress volume → height → speed · every 2–3 days	
☐	**4 · Sport-specific**	Return to training → competition · keep stage 1–2 twice weekly	Stage 4 tolerated with pain ≤ 3/10 → compete

Progressive tendon loading beat eccentric decline squats: VISA-P +28 vs +18 points at 24 wk [M]. Isometrics: analgesia not consistently better than isotonic loading — use for load tolerance, not as the whole programme.

QUADRICEPS STRENGTH — 12-WEEK HEAVY-SLOW PROGRESSION & LOAD LOG

3 SESSIONS / WK

Exercises: ☐ Leg press ☐ Split squat ☐ Leg extension (OKC) ☐ Step-up ☐ Spanish squat ☐ Hamstring curl / RDL

BLOCK	WK 1	WKS 2–3	WKS 4–5	WKS 6–8	WKS 9–12
Target RM × sets	**15RM × 3**	**12RM × 3**	**10RM × 4**	**8RM × 4**	**6RM × 4**
Load (kg)					
Pain during / 24 h					

GREEN 0–2/10 — safe, progress load · **AMBER 3–5/10** — acceptable if settled by next morning · **RED > 5/10** or worse next day — regress one step

BLOOD FLOW RESTRICTION (BFR)

BFR Position Stand 2019 · AOP · DOPPLER

☐ **AOP** _____ mmHg → working **40–80% AOP** = _____ mmHg Calc

☐ **Active BFR:** 30-15-15-15 reps at 20–40% 1RM, 30–60 s rest, 5–10 min per exercise then deflate · 2–3×/wk [M]

⛔ **Cuff on the PROXIMAL THIGH only — never below the knee (common peroneal nerve).**

Exclude if: ☐ DVT / VTE history ☐ PAD ☐ Sickle cell ☐ Uncontrolled HTN ☐ Pregnancy ☐ Infection / malignancy

RETURN-TO-RUNNING GATE (POST-OP)

ALL REQUIRED

- ☐ ≥ 12 weeks after ACLR
- ☐ Effusion ≤ trace
- ☐ Quadriceps LSI > 80%
- ☐ 30 min brisk walk pain-free
- ☐ Full extension, flexion ≥ 95%
- ☐ Pain ≤ 2/10
- ☐ Pain-free repeated SL hops
- ☐ Normal gait, no limp

Expert-consensus criteria [E]; most studies use time alone (median 12 wk) — add the clinical criteria.

COMPREHENSIVE KNEE RETURN-TO-SPORT BATTERY

Delaware-Oslo 2016 · PASS: LSI ≥ 90% UNLESS STATED

TEST	PASS CRITERION	LEFT	RIGHT	LSI %	PASS
STRENGTH					
Quadriceps (isokinetic 60°/s or HHD)	LSI ≥ 90% · pivoting sport aim 100% [M]				☐
Hamstrings	LSI ≥ 90%				☐
HOP BATTERY					
Single hop for distance ▸	LSI ≥ 90% [M]				☐
Triple hop for distance	LSI ≥ 90%				☐
Crossover hop	LSI ≥ 90%				☐
6-m timed hop	LSI = uninvolved ÷ involved time ≥ 90%				☐
POWER, LANDING & BALANCE					
Countermovement jump height	LSI ≥ 90% · drop-jump RSI > 1.3 (2-leg) / > 0.5 (1-leg) field sport [E]				☐
Landing Error Scoring System	≤ 4 errors (≥ 5 = risk cut-point, Sn 86% Sp 64%) [L]	score _____			☐
Y-Balance — anterior reach (cm) ▸	Asymmetry < 4 cm (> 4 cm ≈ 2.5× injury odds) [M]				☐
Change of direction (5-0-5 / T-test, s)	Within pre-injury / team norm · COD deficit symmetric [E]				☐
PATIENT-REPORTED & TIME					
KOS-ADLS / global rating	≥ 90% [M]	_____ % · _____ %			☐
ACL-RSI · TSK-11	Rising trend, no fear-avoidance flags [M]	RSI _____ · TSK _____			☐
Time since ACLR	≥ 9 months — reinjury ↓ 51% per month of delay up to 9 mo [M]	_____ months			☐

RTS decision: ☐ Cleared — full RTS ☐ Modified training ☐ Not cleared **Tests passed** _____ / 13

LSI caution: the uninvolved limb also deconditions, so LSI overestimates recovery — compare with pre-injury or normative values where available [M]. Each 1% gain in quadriceps symmetry ≈ 3% lower reinjury rate.

MOVEMENT QUALITY · LANDING MECHANICS · READINESS NOTES

Longitudinal Session-by-Session Tracking Flowsheet

Patient _____ 09 / 09

#	DATE	NPRS R/A	EFFUSION	EXT / FLEX °	SLR LAG	MODALITIES DELIVERED	MANUAL & NEURO APPLIED	LOAD & STAGE	ADHERENCE	INIT
1						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
2						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
3						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
4						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
5						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
6						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
7						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
8						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
9						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
10						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
11						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		
12						ESWT | PBM | TECAR | BFR | Cryo | NMES | Cup	Maitland | Mulligan | DN | IASTM | Neuro	St 1 | St 2 | St 3 | St 4		

DISCHARGE & MILESTONE SUMMARY

FINAL OUTCOME

PROM	INITIAL	DISCHARGE	Δ	MCID MET
KOOS Sport / Rec (8–10)	/100	/100		☐
IKDC subjective (11.5)	/100	/100		☐
Kujala / VISA-P (10 / trend)	/100	/100		☐
Oxford Knee Score (7)	/48	/48		☐
ACL-RSI (trend)	/100	/100		☐
PSFS average (MDC 2)	/10	/10		☐

Discharge: ☐ Goals met ☐ Self-management plan ☐ Referred on ☐ Non-responder (p4 rule)

OBJECTIVE RE-TEST MILESTONES

L / R

MEASURE	BASE	WK 4	WK 8	WK 12	D/C
Extension °					
Flexion °					
Effusion grade					
Quad LSI %					
Single hop LSI %					
Y-Bal ant. Δ cm					

HOME PROGRAMME · DISCHARGE ADVICE · RELAPSE PLAN

Physician follow-up / sign-off _____ Signature _____ Date _____

Discharging Clinician _____ Sig. _____